"Joe's Hospital of Fried Foods" vs. National Average
T-Test Tutorial for BIOL 318/418



A **T-test** is a statistical test used when you want to know if the average of your sample is meaningfully different from a known value, but you cannot assume the population standard deviation is known. Instead, it uses the sample's own standard deviation and adjusts for smaller sample sizes through the t-distribution. The result is a t test statistic, which tells you how far the sample mean is from the hypothesized population mean in units of estimated standard error. Because it accounts for added uncertainty, the t-test is especially useful when working with small samples (**fewer than 30**) or when the population variability is unknown.

Part 2 Scenario:

The New Challenge: Prove It with a One-Sample T-Test

The awards committee has pointed out that Joe's approach wasn't entirely sound. They raised two key issues:

1. Assumed Population Standard Deviation – Joe can no longer assume the national standard deviation applies, given the variation in practices across hospitals.
2. Small Sample Size – Joe's sample wasn't large enough to justify using a Z-test, which works best with larger samples, N>=30.

The New Scenario: Enter the T-Test

Joe now faces a more realistic challenge. Without knowing the population standard deviation, Joe must rely on his sample's standard deviation to redo the analysis. This is where the T-test comes into play. The T-test

is designed to account for the smaller sample size and the added uncertainty from estimating the population standard deviation.

Feeling the pressure but determined to stay in the running for the Baby Butter Ball Award, Joe opens his spreadsheet once again. This time, he must:

- Calculate the sample mean and the sample standard deviation from his hospital's baby weight data.
- Perform a one-sample T-test to determine if his babies' weights are significantly larger than the national average of 7.5 lbs.

Step 1: Why Use a T-Test Now?

1. **Unknown Population Standard Deviation:** In real-world situations, we often **do not have** the population standard deviation. Therefore, we estimate it using the **sample standard deviation**.
2. **Small Sample Size:** If our sample size is small (e.g., less than 30 newborns), the t-test is preferred because it adjusts for extra variability in the data.

Step 2: Set Up Your Google Sheet for the T-Test

1. **Return to your spreadsheet from the Z-test in Part 1**
2. ***Again***, Copy the data from the first Google Sheet, tab named "***Data for Part 1 and 2***" and paste the data into the sheet tab named **"Part 2: National T-Test"**
3. Working in the tab **"Part 2: National T-Test"**
4. **Columns D and E:** Set aside this column for statistical calculations.
5. In cell **D1**, it should say "**Statistical Names**"
6. In cell **E1**, it should say "**Statistical Values**"
7. In cell **D2**, it should say "Population Mean"
8. In cell **E2**, enter "`7.5`"
    a. ^ this is our representative mean baby weight of the population
9. In cell **D3**, it should say "Population St.Dev."
10. In cell **E3**, leave as not used, since we will not be using it
11. After completion your table should look similar to this:

Statistical Names	Statistical Values
Population Mean	7.5000
Population St.Dev.	not used

Step 3: Calculate the Sample Size

1. In cell **D5**, it should say **"Sample Size."**
2. In cell **E5**, use the `COUNT` function to dynamically count the number of weights:
    - Enter: `=COUNT(B2:B101)`

Step 4: Calculate the Sample Mean

1. In cell **D6**, it should say **"Sample Mean."**

2. In cell **E6**, use the `AVERAGE` function to calculate the sample mean:
    - Enter: `=AVERAGE(B2:B101)`

Step 5: Calculate the Sample Standard Deviation

1. In cell **D7**, it should say **"Sample St.Dev."**
2. In cell **E7**, use the `STDEV.S` function to calculate the sample standard deviation:
    - Enter: `=STDEV.S(B2:B101)`

Step 6: Calculate the T-Statistic

1. In cell **D9**, it should say **"T-Statistic."**
2. In cell **E9**, use this formula to calculate the t-statistic:
    - Enter: `=(E6 - E2) / (E7 / SQRT(E5))`
    - This formula uses the sample mean (E6), the assumed population mean (E2 at 7.5 lbs), the sample standard deviation (E7), and the sample size (E5).

Step 7: Calculate the Degrees of Freedom

1. In cell **D10**, it should say **"Degrees of Freedom."**
2. In cell **E10**, calculate the degrees of freedom using:
    - Enter: `=E5 - 1`
    - Here, `E5` is the sample size, so subtracting 1 gives us the degrees of freedom.

Explanation: Why Do We Use Degrees of Freedom?

- **What Are Degrees of Freedom?** In statistics, **degrees of freedom** represent the number of independent values in a dataset that can vary when estimating a statistical parameter. In the context of a t-test, degrees of freedom are used to account for the fact that we are estimating the population standard deviation using the **sample standard deviation**.
- **Why Subtract 1?** When calculating the **sample standard deviation**, one of the values (usually the mean) is fixed based on the rest of the data. This is why we lose one degree of freedom, which adjusts for the uncertainty introduced when estimating the standard deviation from a limited sample.
- **Impact on the T-Distribution:** The t-distribution changes shape depending on the degrees of freedom. For smaller sample sizes (fewer degrees of freedom), the t-distribution has **heavier tails**, reflecting greater variability and uncertainty. As the sample size grows (increasing degrees of freedom), the t-distribution becomes more similar to the **normal (z) distribution**.
- **Why It Matters:** In the t-test, degrees of freedom help adjust the **critical values** and **p-values** to reflect the sample size's impact. Smaller degrees of freedom make the test more conservative, meaning we need stronger evidence to reject the null hypothesis.

Step 8: Calculate the P-Value for the T-Test

8.1: Understand the Research Question

- **One-Tailed Test:** Use this if you want to test whether the babies at "Joe's Hospital of Fried Foods" are **larger** than the national average. This type of test only looks for a difference in **one direction** (greater than).
- **Two-Tailed Test:** Use this if you want to check for **any significant difference** (either larger or smaller) in baby weights compared to the national average. This type of test considers differences in **both directions**.

8.2: Calculate the P-Value Based on the Chosen Test

1. **Determine the Type of Test:** Decide if you are conducting a **two-tailed test** or a **one-tailed test** based on the research question:
    - **Two-Tailed Test:** Tests for any difference (either larger or smaller) in baby weights.
    - **One-Tailed Test:** Tests if the baby weights are **specifically larger** than the national average.
2. **Calculate the P-Value:**
    - **Two-Tailed Test:**
        1. In cell **D12**, it should say **"P-Value (Two-Tailed)."**
        2. In cell **E12**, use the `T.DIST.2T` function:
            - Enter: `=T.DIST.2T(ABS(E9), E10)`
            - This formula calculates the p-value for a **two-tailed test** using the **absolute value** of the t-statistic (in cell E9) and the **degrees of freedom** (in cell E10).
            - **Explanation:** The `T.DIST.2T` function checks for differences in both directions (greater or smaller) between the sample mean and the population mean.
    - **One-Tailed Test:**
        1. In cell **D13**, it should say **"P-Value (One-Tailed)."**
        2. In cell **E13**, use the `T.DIST.RT` function:
            - Enter: `=T.DIST.RT(E9, E10)`
            - This formula calculates the p-value for a **one-tailed test** using the t-statistic (in cell D5) and the **degrees of freedom** (in cell D6).
            - **Explanation:** The `T.DIST.RT` function tests the probability of observing a value as extreme or more extreme in **one direction** (greater than the population mean).

8.3 Important Notes:

- The **two-tailed p-value** is generally **double** the one-tailed p-value because it considers deviations in both directions.
- Make sure to use the correct function based on whether you are conducting a **one-tailed** or **two-tailed** test. This choice directly impacts the interpretation of your hypothesis test.

Step 9: Decision Using an IF Statement

1. In cell **D15**, it should say "Significance Level (α)."

2. In cell **E15**, enter "`0.05`"
3. In cell **D17**, it should say "*Two-Tail Decision*."
4. In cell **E17**, use the **IF** statement to automatically determine whether to reject the null hypothesis:
    - Using the **two-tailed test** p-value (in **E12**), enter the formula:
      `=IF(E12 <= E15, "Reject the Null Hypothesis", "Fail to Reject the Null Hypothesis")`
5. In cell **D18**, it should say "*One-Tail Decision*."
6. In cell **E18**, use the **IF** statement to automatically determine whether to reject the null hypothesis:
    - Using the **one-tailed test** p-value (in **E13**), enter the formula:
      `=IF(E13 <= E15, "Reject the Null Hypothesis", "Fail to Reject the Null Hypothesis")`

Step 10: Interpret the Results

1. In cell **D20**, it should say **"Interpretation."**
2. In cell **E20**, write a brief interpretation of your findings based on the p-value and test type.

Practice Task:

- Change some baby weights in Column B and observe how the sample standard deviation, t-statistic, and p-value adjust automatically.
- Add more baby weights (e.g., increasing the sample size beyond 30) and note how the p-value changes. Discuss when the t-test results start to resemble the z-test.
- Discuss how the flexibility in sample size affects the results of the t-test and compare this to their earlier z-test results.

Key Takeaways:

1. Use the **t-test** when the **population standard deviation is unknown** and/or the **sample size is small**.
2. The **t-distribution** accounts for the additional uncertainty introduced by using the sample standard deviation.
3. When the sample size is large, the t-test and z-test produce similar results, showing that the t-distribution approximates the normal distribution as the sample size increases.

Joe's Redemption – Reinstating the Award

After carefully following the steps, Joe anxiously awaits the results. To his relief, the T-test shows that his babies are indeed significantly larger than the national average. With this, the committee reinstates his eligibility for the Baby Butter Ball Award.

As Joe accepts the news, he feels a sense of relief but also humility. "I've learned a valuable lesson," he admits, holding back his excitement. "From now on, I'll make sure I'm using the right statistical tools."

But Joe can't resist a small smile. For now, the award is within his grasp again, and he's more determined than ever to prove that fried food really does make babies thrive… or at least win awards.